

Resonant Inelastic X-ray Scattering by Dynamical Mean-Field Theories

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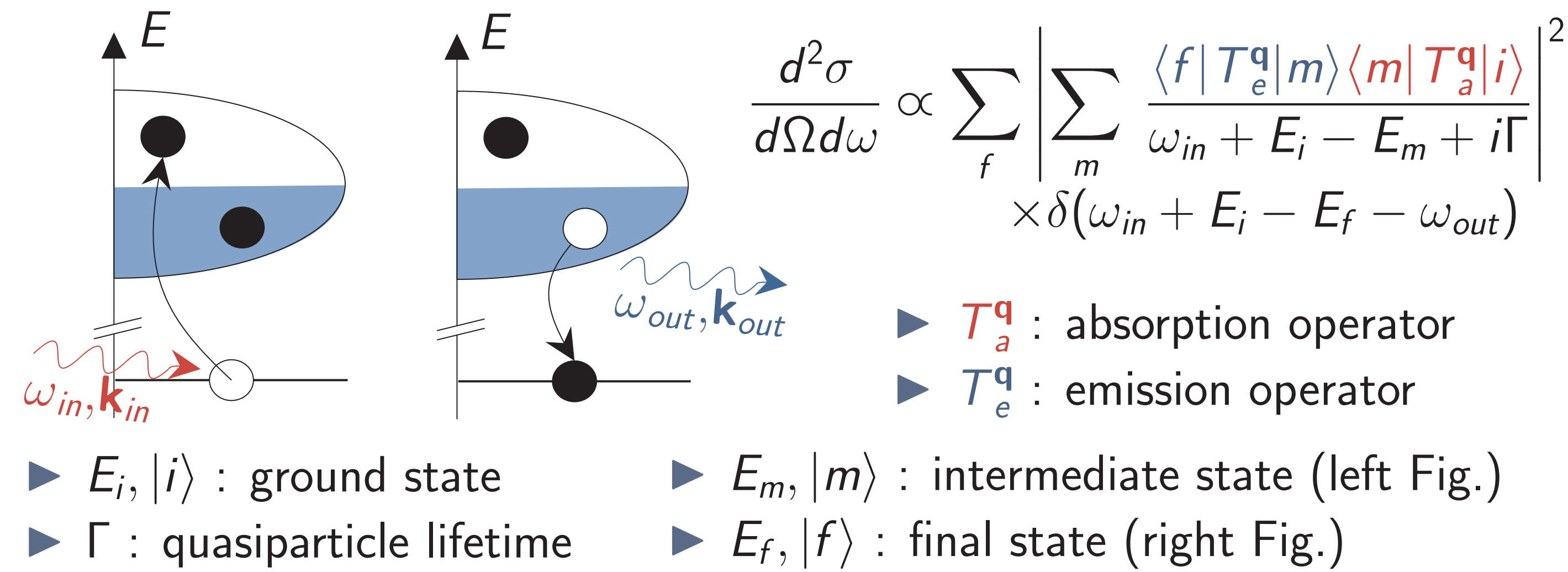
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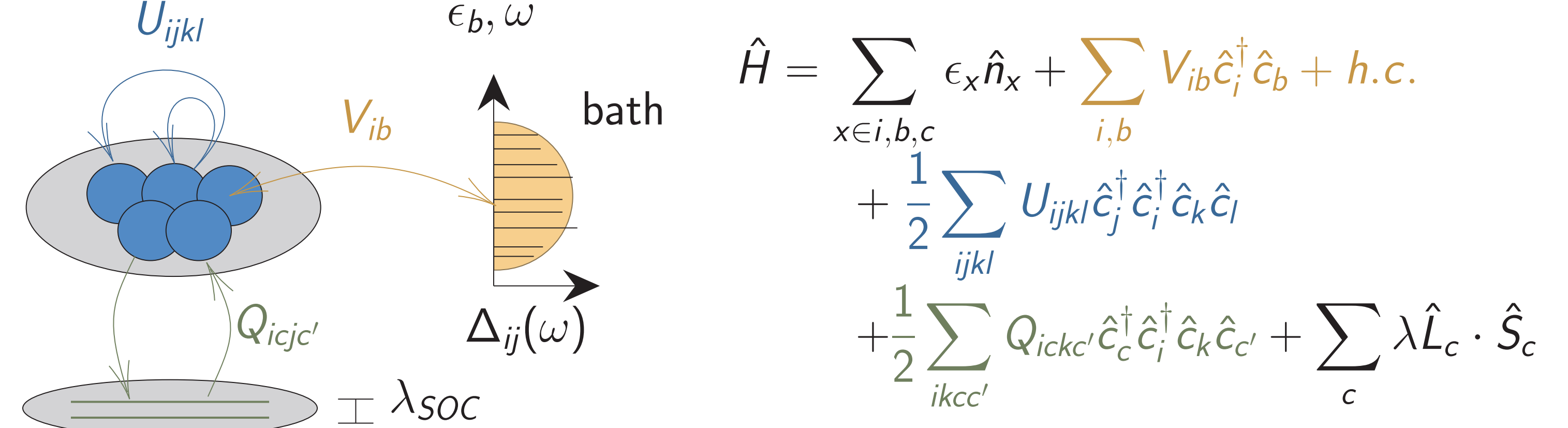
Resonant Inelastic X-ray Scattering (RIXS)

RIXS is a two process spectroscopy technique with core **absorption** and **emission**. It allows to probe a lot of different excitations (phonons, magnons, $d-d$, charge transfer) via the cross section σ .



Augmented Anderson Impurity Model (AAIM)

The model of interest is the AAIM, corresponding to an impurity with **core** and **valence** states, the latter hybridizing with a **non-interacting bath**.



Building the model

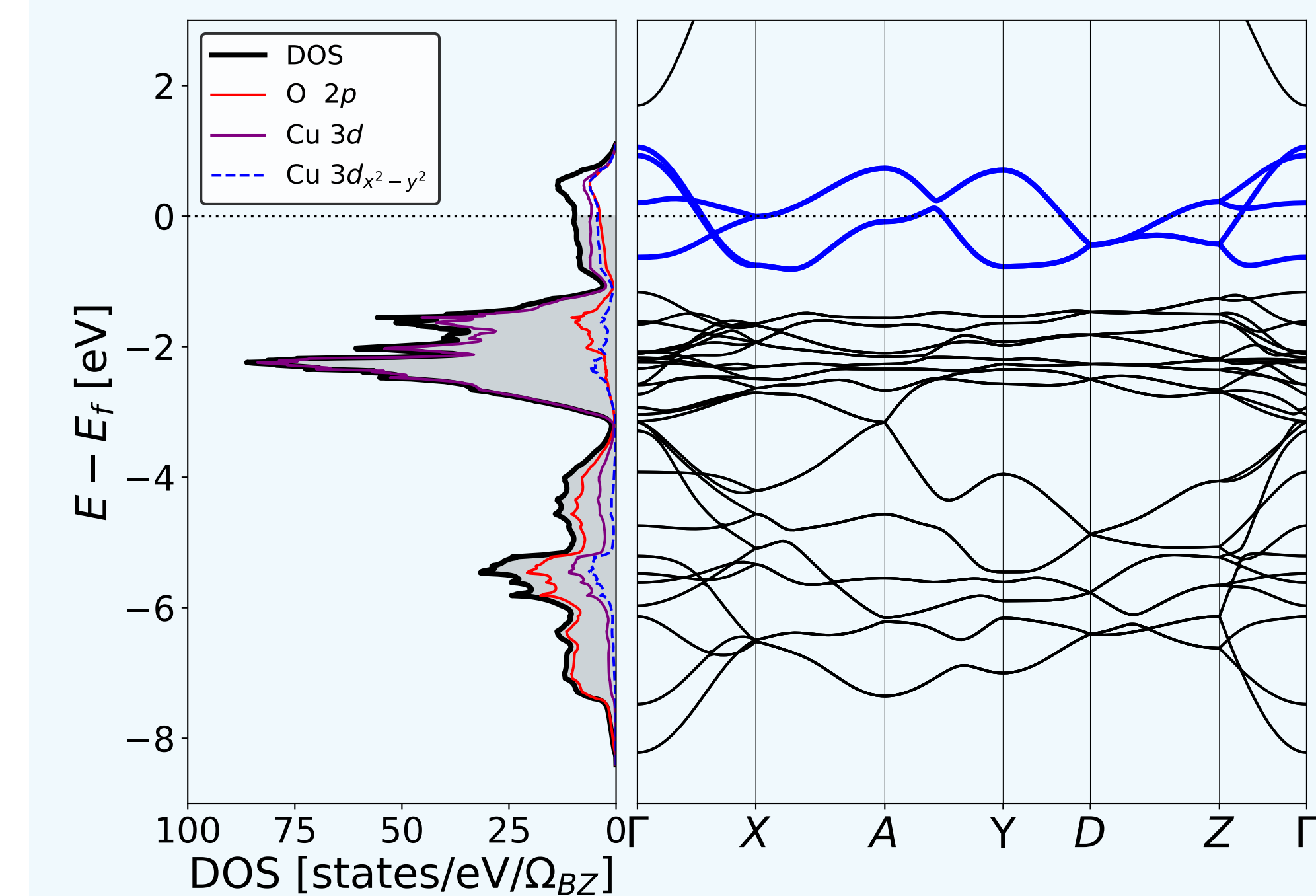
- Density Functional Theory
- Dynamical Mean Field Theory

Solving the model

- Matrix Product State formalism
- Exact Diagonalization (Lanczos)

Density Functional Theory (DFT)

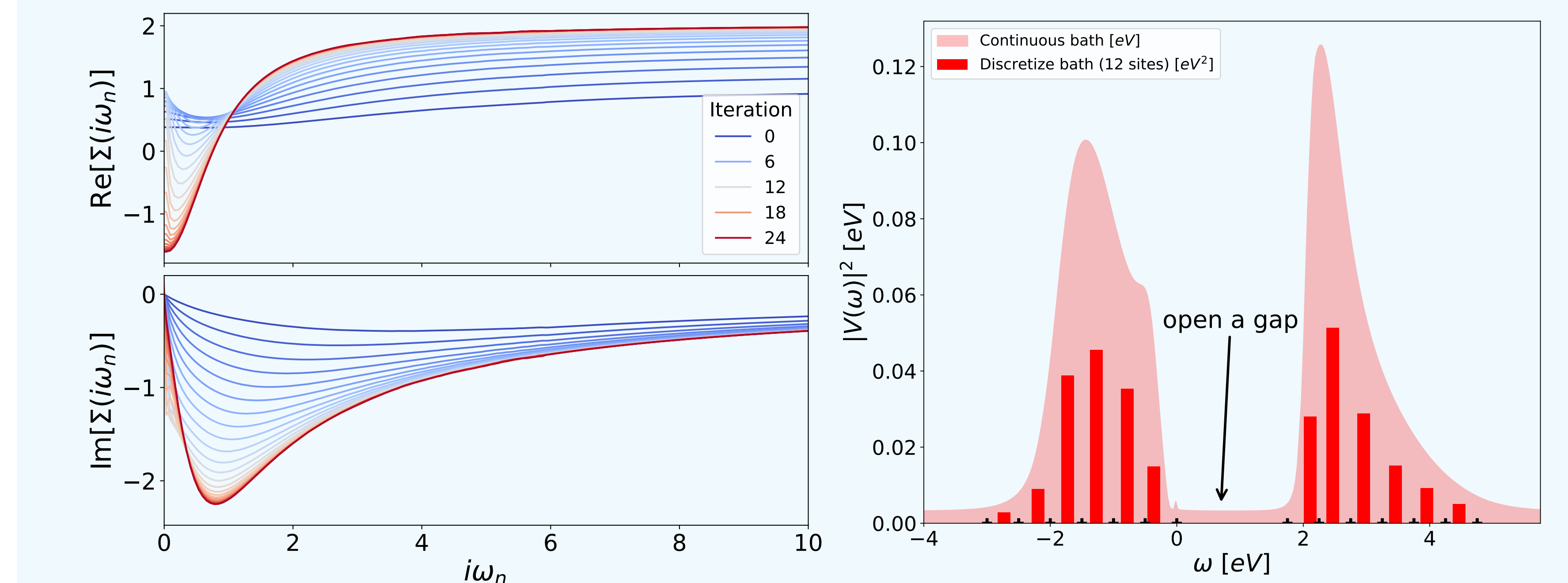
One band model benchmark with copper oxide CuO.



- Metallic phase** although experiments measure a gap.
- Copper $d_{x^2-y^2}$'s hybridize with oxygen 2p's at E_f .

Dynamical Mean Field Theory (DMFT)

DMFT maps the lattice problem self-consistently onto an impurity model, solved with Monte-Carlo.



- Extract the converged self-energy $\Sigma(i\omega_n)$ and **continue it analytically** to real frequencies.
- Compute the continuous bath $|V(\omega)|^2 \propto \Im(\Delta(\omega)) = G^{-1}(\omega) + \Sigma(\omega)$ and **discretize it**.

Matrix Product States (MPS)

A many-body state $|\psi\rangle$ on a basis $|\mathbf{n} = n_1 \dots n_L\rangle$:

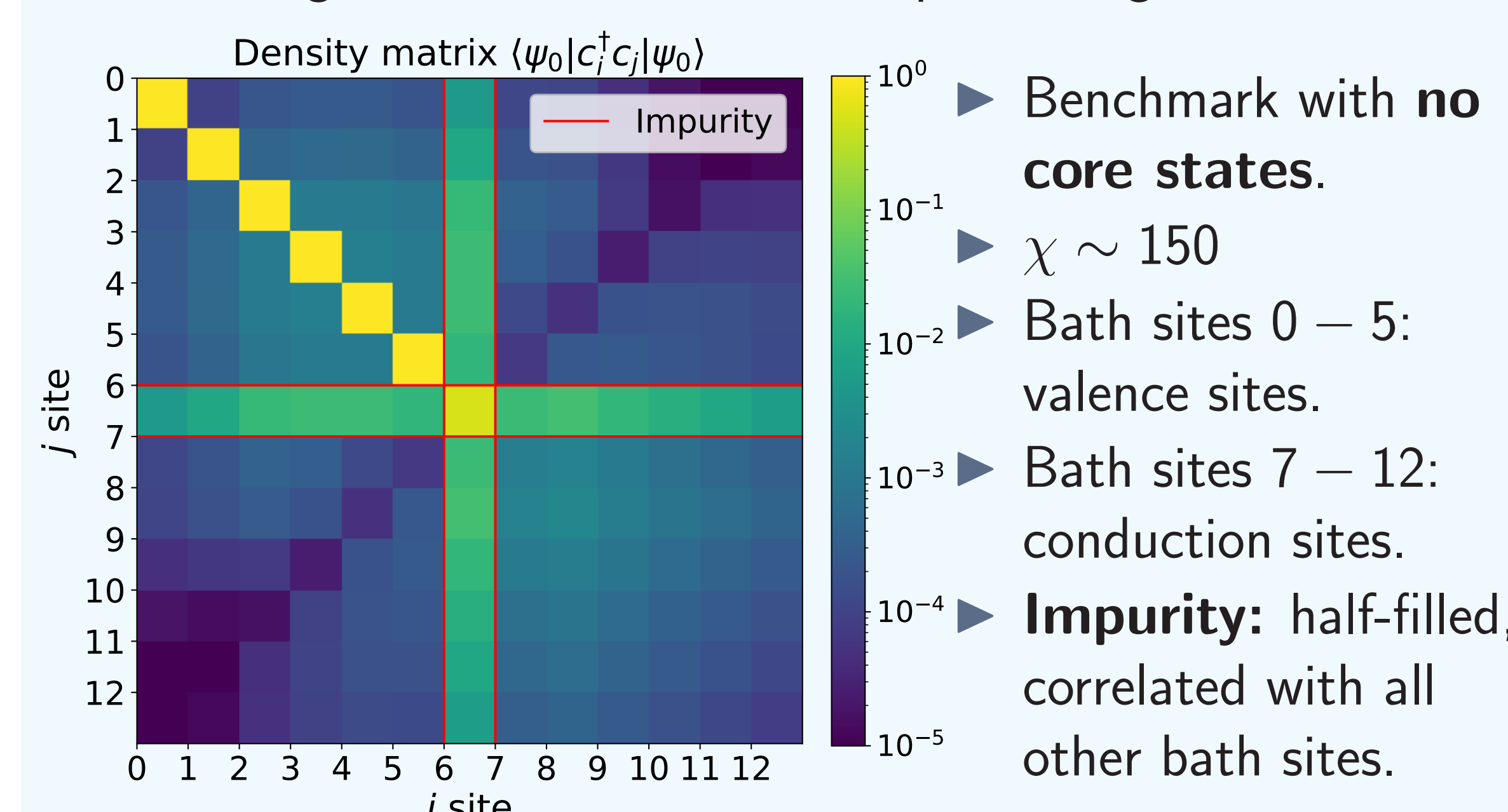
$$|\psi\rangle = \sum_{\mathbf{n}} c_{\mathbf{n}} |\mathbf{n}\rangle = \sum_{\mathbf{n}} \sum_{a_1 \dots a_{L-1}} M_{1,a_1}^{n_1} M_{a_1,a_2}^{n_2} \dots M_{a_{L-1},1}^{n_L} |\mathbf{n}\rangle$$

Diagram illustrating the MPS representation. A chain of rank-3 tensors M^i is shown, with indices n_i and a_i .

- MPS = chain of rank-3 tensors M^i with dimension that grows exponentially with L .
- Truncating the singular values:** $L\chi^2$ complexity while keeping (hopefully) the interesting physics!
- χ , that is the number of singular values kept, control this compression.

Density Matrix Renormalization Group

Density Matrix Renormalisation Group (DMRG) is a variational method using MPS that allows to compute the ground state.



- Benchmark with **no core states**.
- $\chi \sim 150$
- Bath sites 0 – 5: valence sites.
- Bath sites 7 – 12: conduction sites.
- Impurity:** half-filled, correlated with all other bath sites.

Lanczos Algorithm

Projection of $\hat{H}$ in the Krylov subspace:

$$\mathcal{K}_k(\hat{H}, v) = \text{span}(v, \hat{H}v, \dots, \hat{H}^k v)$$

- $v = T_a^q |i\rangle$ the initial state
- $k \sim 10^6$ the dimension of the subspace where diagonalisation is still manageable.
- Lanczos orthogonalizes the states iteratively.

Problem: applying $\hat{H}$ on a state increases the bond dimension χ significantly and thus the computational complexity...

Natural Orbitals

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Conclusion and Outlook

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References

- [1] H. Yavas, M. Sundermann, K. Chen, A. Amorese, A. Severing, H. Gretarsson, M. W. Haverkort, and L. H. Tjeng. *Direct imaging of orbitals in quantum materials*. *Nature Physics*, 15(6):559–562, 2019.
- [2] Right figure of the *Motivation* box : experimental results of orbital imaging of copper oxide at 300 K, by Victor Baledent and Jean-Pascal Rueff, with whom we are collaborating. The NRIXS spectra comes from paper [1] for NiO.
- [3] D. Tahir and S. Tougaard. *Electronic and optical properties of Cu, CuO and Cu2O studied by electron spectroscopy*. *Journal of Physics: Condensed Matter*, 24(17):175002, apr 2012.
- [4] F. Marabelli, G. B. Parravicini, and F. Salghetti-Drioli. *Optical gap of CuO*. *Phys. Rev. B*, 52:1433–1436, Jul 1995.
- [5] We gratefully acknowledge financial support from the ANR ClusterSpecs project.